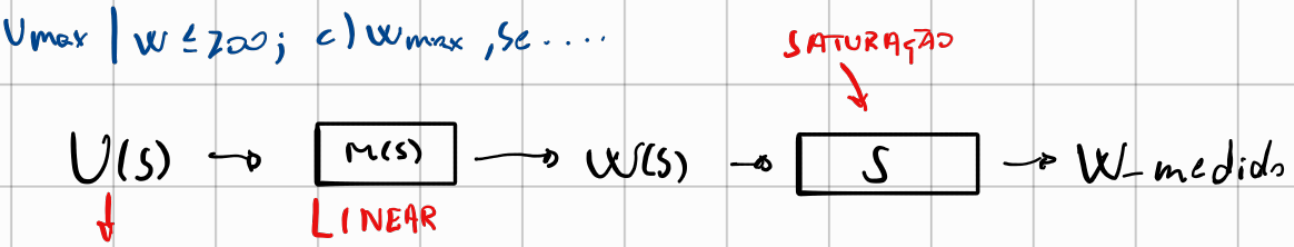
b) U_{max} | $w \leq 200$; c) w_{max}, se ....



-50 a 50

$U(s) = \frac{R}{s}$

BEGIN

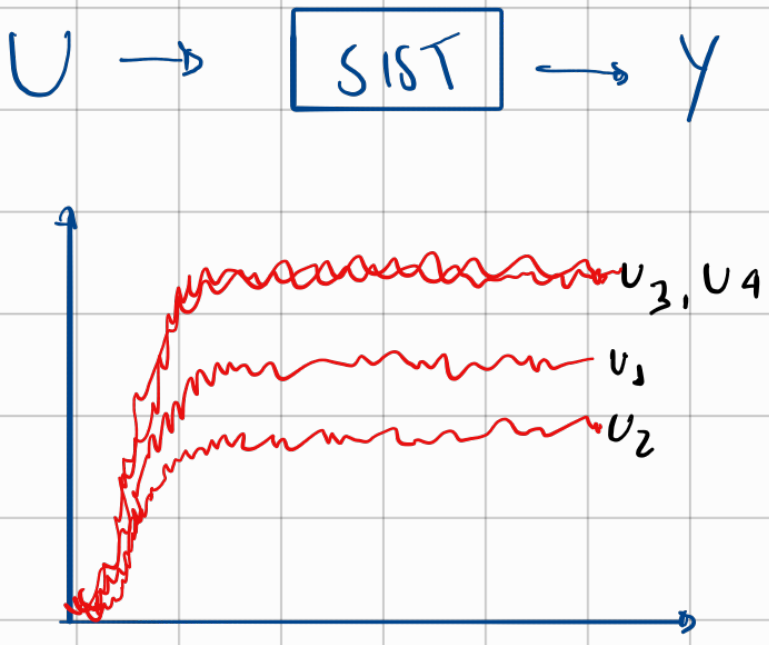
$y = m(t) \cdot G$

if $y \geq 200$

$y = 200$

endif

END



$U_4 > U_3$

$$y(t) = \theta_3 + \theta_2 \cdot e^{-\theta_3 t} + \theta_4 e^{-\theta_5 t}$$

$$\Downarrow \mathcal{L}\{y(t)\}$$

$$Y(s) = \frac{\theta_3}{s} + \theta_2 \cdot \frac{1}{s+\theta_3} + \theta_4 \cdot \frac{1}{s+\theta_5}$$

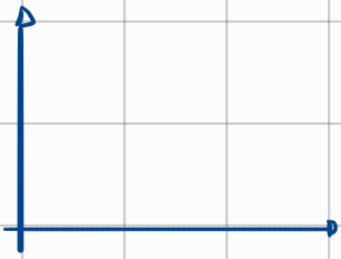
$$\begin{cases} \theta_1 = 100,02 \\ \theta_2 = 27.85 \\ \theta_3 = 58,43 \\ \theta_4 = -126,31 \\ \theta_5 = 9,49 \end{cases}$$

$$M(s) = \frac{\theta_1\theta_3\theta_5/K}{(s+\theta_3)(s+\theta_5)} = \frac{\theta_1\theta_3\theta_5/K}{s^2 + (\theta_1+\theta_3)s + \theta_1\theta_3}$$

$$M(s) = \frac{11100}{s^2 + 67.9s + 555}$$

TVF:

$$y(t)\big|_{t\to\infty} = Y(s)\cdot s\big|_{s\to 0} = \frac{11100}{s^2+67.9s+555}\cdot\frac{K}{s}\cdot s\Big|_{s\to 0}$$



$$y(t)\big|_{t\to\infty} = \frac{11100}{555}\cdot K = 20\cdot K$$

$$y_{max} = 200 = 20K$$

$$K = 10$$

W_{max} ocorre quando U_{max} $\Gamma U_{max} = 50$

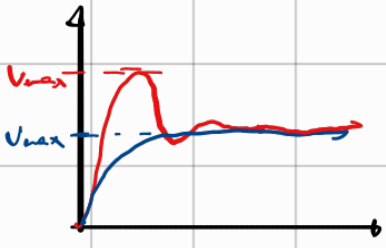
$$\mathcal{L}\,u(t) = \frac{K}{s} = \frac{50}{s}$$

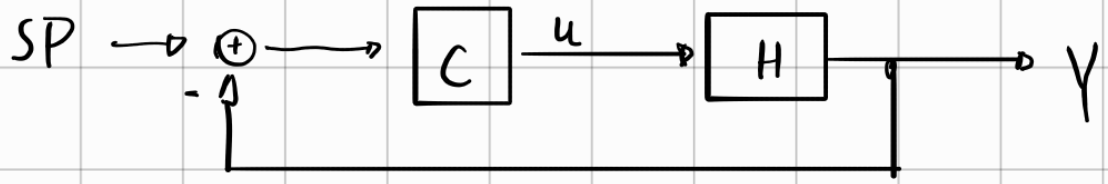
$$y_{max} = 200 = 20K = 20\cdot 50 = 1000 \text{ rad/s}$$

TVF $\rightarrow$ $y_{est} = 20\cdot K$

$$y_{MAX} = 20\cdot U_{MAX}$$

$$y_{MAX} = 20\cdot 50 = 1000$$





$$Y = \left(\frac{K}{S} \cdot e\right) H$$

$$H = \frac{Y \cdot S}{K \cdot e}$$

$$\frac{Y}{SP} = \frac{CH}{1+CH}$$

$$\frac{Y}{SP}$$

$$G = \frac{CH}{1 + CH}$$

$$H = \frac{\theta_1\theta_3\theta_5/K}{(S+\theta_3)(S+\theta_5)} =$$

$$\frac{0{,}0098}{S^2 + 0{,}256S + 0{,}0098} = \frac{Y(S)}{U(S)}$$

(0,0098 → a; 0,256 → b; 0,0098 → a)

$$Ys^2 + bYt + aY = aU$$

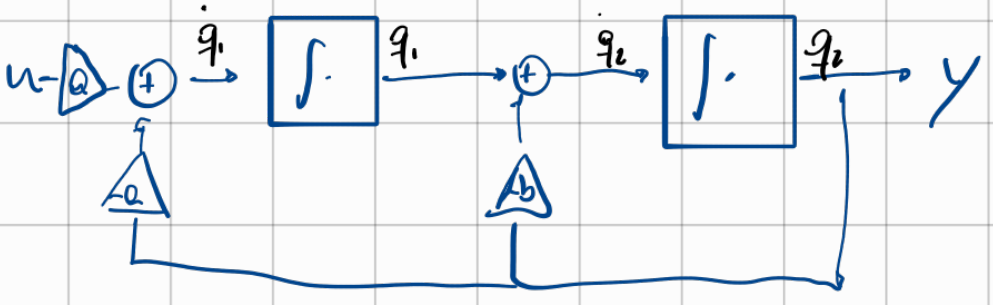
$$\Updownarrow \mathcal{L}^{-1}$$

$$\ddot{y} + b\dot{y} + ay = a \cdot U$$

$$\Updownarrow$$

$$y - b\int y\,dt - a\iint y\,dt = a\iint u\,dt$$

$$y = a\iint u - a\iint y - b\int y$$



$$\dot{q}_1 = a \cdot u - a q_2$$

$$\dot{q}_2 = q_1 - b q_2$$

$$y = q_2$$

$$\dot{q} = Aq + Bu$$

$$y = Cq + Du$$

$$A = \begin{bmatrix} 0 & -a \\ 1 & -b \end{bmatrix} \; ; \; B = \begin{bmatrix} a \\ 0 \end{bmatrix}$$

$$C = [0 \;\; 1] \; ; \; D = 0$$

$$G = \frac{CH}{1 + CH}$$

$$G = \frac{\frac{K}{s} \cdot \frac{a}{(s+\theta_3)(s+\theta_5)}}{1 + \frac{K}{s} \cdot \frac{a}{(s+\theta_3)(s+\theta_5)}}$$

$$\frac{Y}{U} = \frac{K \cdot a}{s^3 + s^2(\theta_3+\theta_5) + s\,\theta_3\theta_5 + K \cdot a}$$

$$H = \frac{a}{s^2 + bs + c} = \frac{Y(s)}{U(s)}$$

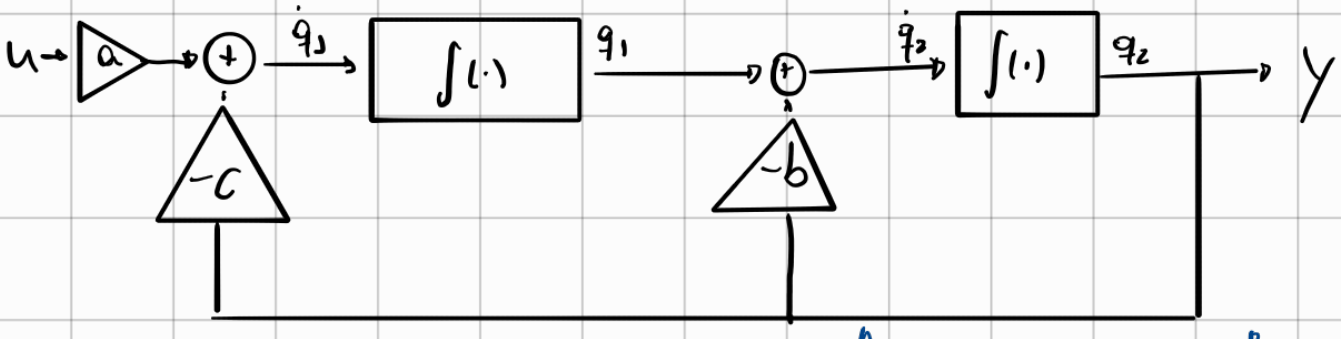
$$Y(s) \cdot s^2 + Y(s) \cdot b \cdot s + Y(s) \cdot c = U(s) \cdot a$$

$$\Updownarrow$$

$$\ddot{y}(t) + b \cdot \dot{y}(t) + c \cdot y = a \cdot u(t)$$

$$y + b \cdot \int y + c \cdot \iint y = a \cdot \iint u$$

$$y = a \iint u - c \iint y - b \int y$$



$$\dot{q}_1 = a \cdot u - c \cdot q_2$$

$$\dot{q}_2 = q_1 - b \cdot q_2$$

$$y = q_2$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -c \\ 1 & -b \end{bmatrix}}_{A} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \underbrace{\begin{bmatrix} a \\ 0 \end{bmatrix}}_{B} [u]$$

$$[y] = \underbrace{[0 \;\; 1]}_{C} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \underbrace{[0]}_{D} [u]$$

$$\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} = \begin{bmatrix} 0 & -c \\ x_3 & x_4 \end{bmatrix} \begin{bmatrix} q_1 \\ q_2 \end{bmatrix} + \begin{bmatrix} a \\ h_2 \end{bmatrix} u$$

$$\dot{q}_1 = x_1 q_1 + x_2 q_2 + h_1 \cdot u$$